Effect of Dropout on Overfitting in Neural Networks

Introduction

Neural networks are powerful models used for classification and prediction tasks. However, they often suffer from overfitting, where the model performs well on training data but poorly on unseen data. This tutorial explores how dropout helps reduce overfitting and improves model generalization.

```
pip install tensorflow matplotlib numpy
```

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Requirement already satisfied: tensorflow in /usr/local/lib/python3.12/dist-packages (2.19.0)
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```

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```

Dataset Description

The MNIST dataset is used in this experiment. It consists of handwritten digit images (0–9). Each image is 28x28 pixels and is commonly used for testing machine learning models.

```
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout
from tensorflow.keras.utils import to_categorical
```

```
# Load dataset
(x_train, y_train), (x_test, y_test) = mnist.load_data()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
```

```
# Normalize data
x_train = x_train / 255.0
x_test = x_test / 255.0

# Flatten images
x_train = x_train.reshape(-1, 28*28)
x_test = x_test.reshape(-1, 28*28)

# One-hot encoding
y_train = to_categorical(y_train)
y_test = to_categorical(y_test)
```

Model Without Dropout

This model does not use dropout. It is expected to overfit the training data due to its capacity to memorize patterns.

```
model1 = Sequential([
    Dense(128, activation='relu', input_shape=(784,)),
    Dense(64, activation='relu'),
    Dense(10, activation='softmax')
])
```

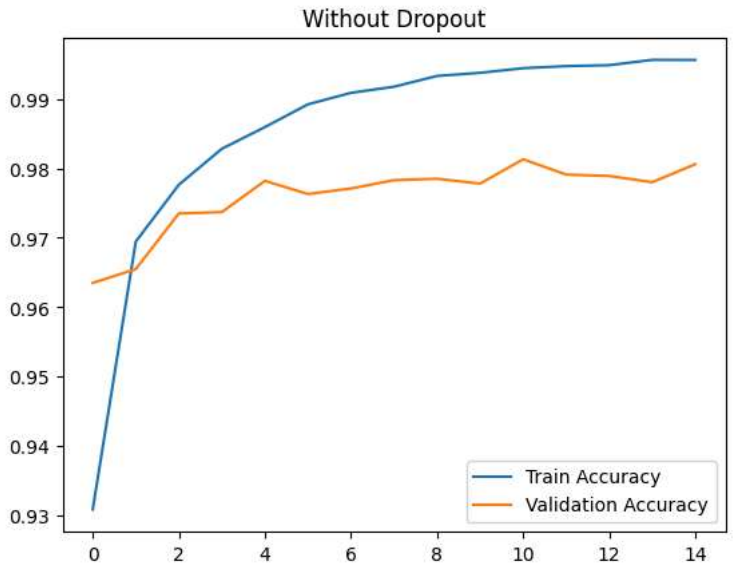
```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential mo
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
model1.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
```

```
history1 = model1.fit(
    x_train, y_train,
    epochs=15,
    validation_data=(x_test, y_test)
)
```

```
Epoch 1/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 11s 5ms/step - accuracy: 0.9309 - loss: 0.2347 - val_accuracy: 0.9635 - val_loss: 0.1165
Epoch 2/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 9s 4ms/step - accuracy: 0.9694 - loss: 0.0997 - val_accuracy: 0.9655 - val_loss: 0.1137
Epoch 3/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9776 - loss: 0.0718 - val_accuracy: 0.9735 - val_loss: 0.0846
Epoch 4/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9828 - loss: 0.0543 - val_accuracy: 0.9737 - val_loss: 0.0847
Epoch 5/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9859 - loss: 0.0443 - val_accuracy: 0.9782 - val_loss: 0.0743
Epoch 6/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9892 - loss: 0.0338 - val_accuracy: 0.9763 - val_loss: 0.0797
Epoch 7/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9909 - loss: 0.0285 - val_accuracy: 0.9771 - val_loss: 0.0925
Epoch 8/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9918 - loss: 0.0255 - val_accuracy: 0.9783 - val_loss: 0.0799
Epoch 9/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9933 - loss: 0.0204 - val_accuracy: 0.9785 - val_loss: 0.0828
Epoch 10/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9937 - loss: 0.0192 - val_accuracy: 0.9778 - val_loss: 0.0907
Epoch 11/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9944 - loss: 0.0176 - val_accuracy: 0.9813 - val_loss: 0.0887
Epoch 12/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9947 - loss: 0.0153 - val_accuracy: 0.9791 - val_loss: 0.1012
Epoch 13/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 3ms/step - accuracy: 0.9948 - loss: 0.0161 - val_accuracy: 0.9789 - val_loss: 0.1099
Epoch 14/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9956 - loss: 0.0132 - val_accuracy: 0.9780 - val_loss: 0.1109
Epoch 15/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9956 - loss: 0.0126 - val_accuracy: 0.9806 - val_loss: 0.1072
```

```
plt.plot(history1.history['accuracy'], label='Train Accuracy')
plt.plot(history1.history['val_accuracy'], label='Validation Accuracy')
plt.legend()
plt.title("Without Dropout")
plt.show()
```



Observation (Without Dropout)

The training accuracy increases significantly, while the validation accuracy is lower. This gap indicates overfitting, where the model memorizes training data but does not generalize well.

Model With Dropout

Dropout is applied to randomly disable neurons during training. This helps reduce overfitting and improves generalization.

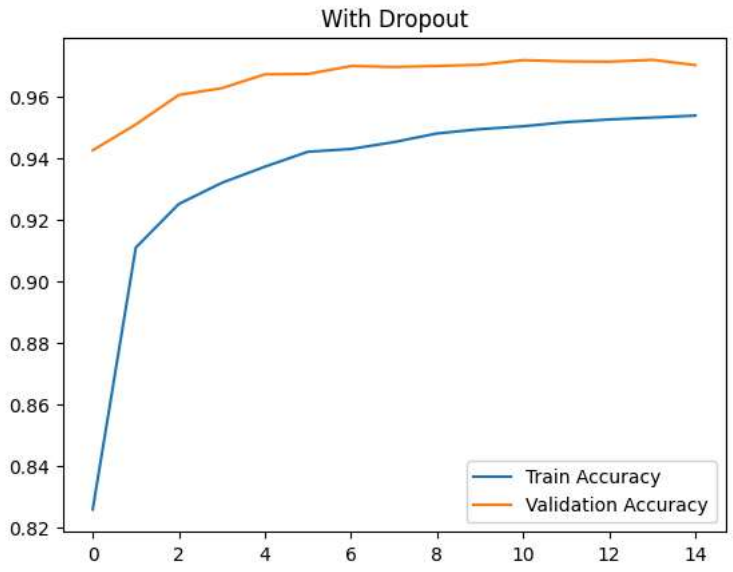
```
model2 = Sequential([
    Dense(128, activation='relu', input_shape=(784,)),
    Dropout(0.5),
    Dense(64, activation='relu'),
    Dropout(0.5),
    Dense(10, activation='softmax')
])
```

```
model2.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
```

```
history2 = model2.fit(
    x_train, y_train,
    epochs=15,
    validation_data=(x_test, y_test)
)
```

```
Epoch 1/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 9s 4ms/step - accuracy: 0.8258 - loss: 0.5674 - val_accuracy: 0.9425 - val_loss: 0.1932
Epoch 2/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9109 - loss: 0.3192 - val_accuracy: 0.9509 - val_loss: 0.1632
Epoch 3/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 9s 4ms/step - accuracy: 0.9251 - loss: 0.2705 - val_accuracy: 0.9605 - val_loss: 0.1324
Epoch 4/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9319 - loss: 0.2410 - val_accuracy: 0.9627 - val_loss: 0.1263
Epoch 5/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9372 - loss: 0.2250 - val_accuracy: 0.9672 - val_loss: 0.1127
Epoch 6/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9420 - loss: 0.2089 - val_accuracy: 0.9673 - val_loss: 0.1120
Epoch 7/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 11s 4ms/step - accuracy: 0.9429 - loss: 0.1988 - val_accuracy: 0.9699 - val_loss: 0.1038
Epoch 8/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9452 - loss: 0.1928 - val_accuracy: 0.9696 - val_loss: 0.1049
Epoch 9/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9480 - loss: 0.1843 - val_accuracy: 0.9699 - val_loss: 0.0992
Epoch 10/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9494 - loss: 0.1788 - val_accuracy: 0.9703 - val_loss: 0.1002
Epoch 11/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9503 - loss: 0.1730 - val_accuracy: 0.9718 - val_loss: 0.0974
Epoch 12/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9517 - loss: 0.1689 - val_accuracy: 0.9714 - val_loss: 0.1036
Epoch 13/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 7s 4ms/step - accuracy: 0.9525 - loss: 0.1674 - val_accuracy: 0.9713 - val_loss: 0.1069
Epoch 14/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9531 - loss: 0.1637 - val_accuracy: 0.9719 - val_loss: 0.1040
Epoch 15/15
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 8s 4ms/step - accuracy: 0.9538 - loss: 0.1602 - val_accuracy: 0.9702 - val_loss: 0.1044
```

```
plt.plot(history2.history['accuracy'], label='Train Accuracy')
plt.plot(history2.history['val_accuracy'], label='Validation Accuracy')
plt.legend()
plt.title("With Dropout")
plt.show()
```



Observation (With Dropout)

The gap between training and validation accuracy is reduced. This indicates that the model generalizes better and overfitting is minimized. The validation accuracy may even be slightly higher due to dropout being active only during training.

Comparison

Model	Training Accuracy	Validation Accuracy	Observation
Without Dropout	High	Lower	Overfitting
With Dropout	Moderate	High	Better generalization

Effect of Different Dropout Values

- Dropout (0.2): Slight overfitting may still occur
- Dropout (0.5): Best balance between training and validation performance
- Dropout (0.8): Model may underfit due to excessive information loss

Conclusion

Dropout is an effective regularization technique that reduces overfitting in neural networks. By randomly disabling neurons during training, it forces the model to learn more generalized features. However, choosing an appropriate dropout rate is important, as too much dropout can lead to underfitting.